

AAK Academy Management System (AAMS)

Database Design Document

V 1.0

By

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CHAPTER 1 PROJECT OVERVIEW

1.1 INTRODUCTION

AAK Academy Management System (AAMS) is a centralized educational management system developed to manage academic and administrative activities efficiently. The system provides a single platform for students, teachers, and administrators.

The system automates important academy operations such as student registration, subject enrollment, assignment submission, quiz handling, attendance management, fee management, live classes, announcements, and notifications.

Traditional academy systems mostly use manual record keeping which creates problems such as data redundancy, inaccurate records, delayed communication, and difficulty in managing academic activities. AAMS solves these problems by providing a centralized database-driven solution.

The system contains three main user roles:

- Student
- Teacher
- Administrator

Students can enroll in subjects, attend live classes, attempt quizzes, submit assignments, and manage fee payments. Teachers can upload learning materials, create assignments and quizzes, and conduct online classes. Administrators manage users, academic programs, and fee structures.

The database design uses advanced concepts including:

- Supertype and subtype relationships
- Associative entities
- Crow's Foot cardinalities
- Normalization
- Participation constraints

These concepts help in reducing redundancy and improving consistency.

1.2 PROBLEM STATEMENT

Many educational academies still use manual systems to manage student records, attendance, assignments, fee collection, and communication. These systems are difficult to maintain and often produce inaccurate records.

Major problems in manual systems include:

- Difficulty in maintaining student records
- Inefficient attendance management
- Delayed communication between teachers and students
- Difficulty in managing assignments and quizzes
- Complicated fee management
- Data redundancy and inconsistency
- Lack of centralized academic management

To solve these problems, AAK Academy Management System (AAMS) is proposed as a centralized database solution.

1.3 PROJECT OBJECTIVES

General Objective

The main objective of AAK Academy Management System is to develop a centralized database system for managing academic and administrative activities efficiently.

Specific Objectives

- To manage student, teacher, and administrator records
- To automate subject enrollment and academic management
- To provide online assignment and quiz management
- To manage live classes and attendance records
- To maintain fee records and payments
- To allow teachers to upload learning materials
- To provide notifications for academic activities
- To maintain audit logs for security and monitoring
- To reduce data redundancy using normalization
- To improve communication between users

1.4 GITHUB REPOSITORY LINK

<https://github.com/bscs24f22-eng/DataBase-Project.git>

CHAPTER 2 DETAILED DATABASE DESIGN

2.1 ENTITY

Sr. No	Entity Name	Description
01	User	Represents all registered users of the system.
02	Student	Represents students enrolled in programs.
03	Teacher	Represents teachers responsible for academic activities.
04	Admin	Represents administrators of the system.
05	Program	Represents academic programs offered by the academy.
06	Subject	Represents subjects offered under programs.
07	Student_Subject	Represents subject enrollment of students.
08	Learning_Material	Represents learning resources uploaded by teachers.
09	Assignment	Represents assignments created by teachers.
10	Assignment_Submission	Represents assignment submissions made by students.
11	Quiz	Represents quizzes created for subjects.
12	Quiz_Question	Represents questions included in quizzes.
13	Quiz_Attempt	Represents quiz attempts made by students.
14	Live_Class	Represents online live classes.
15	Attendance	Represents attendance records of students.
16	Fee_Record	Represents fee details and invoices.
17	Payment	Represents fee payments made by students.
18	Announcement	Represents announcements shared by teachers.
19	Notification	Represents notifications sent to users.
20	Audit_Log	Represents activity logs for monitoring and security.

2.2 DATA DICTIONARY

2.2.1 User

Sr. No	Name	Data Type	Constraint	Description
01	user_id	Integer	Primary Key	Unique ID of user
02	full_name	Varchar(100)	NOT NULL	Full name of user
03	email	Varchar(100)	UNIQUE	Email address
04	password_hash	Varchar(255)	NOT NULL	Encrypted password
05	role	Varchar(20)	NOT NULL	User role
06	phone	Varchar(20)	NULL	Contact number
07	status	Varchar(20)	NOT NULL	Account status
08	created_at	Datetime	NOT NULL	Account creation date
09	last_login	Datetime	NULL	Last login time

2.2.2 Student

Sr. No	Name	Data Type	Constraint	Description
01	student_id	Integer	Primary Key	Unique student ID
02	user_id	Integer	Foreign Key	References User
03	registration_no	Varchar(50)	UNIQUE	Registration number
04	dob	Date	NOT NULL	Date of birth
05	gender	Varchar(10)	NOT NULL	Gender of student
06	address	Text	NULL	Student address
07	enrollment_date	Date	NOT NULL	Enrollment date
08	current_status	Varchar(20)	NOT NULL	Academic status
09	program_id	Integer	Foreign Key	References Program

2.2.3 Teacher

Sr. No	Name	Data Type	Constraint	Description
01	teacher_id	Integer	Primary Key	Unique teacher ID
02	user_id	Integer	Foreign Key	References User
03	qualification	Varchar(100)	NOT NULL	Qualification
04	specialization	Varchar(100)	NOT NULL	Area of specialization
05	experience_years	Integer	NOT NULL	Teaching experience
06	hire_date	Date	NULL	Hiring date

2.2.4 Admin

Sr. No	Name	Data Type	Constraint	Description
01	admin_id	Integer	Primary Key	Unique admin ID
02	user_id	Integer	Foreign Key	References User
03	access_level	Varchar(50)	NOT NULL	Access level
04	permissions	Text	NULL	Admin permissions

2.2.5 Program

Sr. No	Name	Data Type	Constraint	Description
01	program_id	Integer	Primary Key	Unique program ID
02	program_name	Varchar(100)	NOT NULL	Program name
03	level	Varchar(50)	NOT NULL	Program level
04	duration	Integer	NOT NULL	Program duration
05	description	Text	NULL	Program description

2.2.6 Subject

Sr. No	Name	Data Type	Constraint	Description
01	subject_id	Integer	Primary Key	Unique subject ID
02	teacher_id	Integer	Foreign Key	References Teacher
03	program_id	Integer	Foreign Key	References Program
04	subject_name	Varchar(100)	NOT NULL	Subject name
05	subject_code	Varchar(20)	UNIQUE	Subject code
06	fee	Decimal	NOT NULL	Subject fee
07	credit_hours	Integer	NOT NULL	Credit hours
08	prerequisite	Varchar(100)	NULL	Required prerequisite

2.3 BUSINESS RULES

- A student must belong to exactly one program.
- A teacher can teach multiple subjects.
- A subject belongs to one academic program.
- Students can enroll in multiple subjects.
- The many-to-many relationship between Student and Subject is resolved using Student_Subject.
- A quiz can contain multiple question types.
- Objective quiz questions can be auto-graded.

- Attendance is automatically marked from Zoom participation.
- Students can make multiple payments against one fee invoice.
- Notifications can be sent to registered users.
- Important user activities are stored in Audit_Log.
- Automatic discounts are applied based on enrolled subjects:
 - 10% discount for 3 subjects
 - 20% discount for 6 subjects

2.4 ENTITY RELATIONSHIP DIAGRAM

The following diagram represents the EERD of AAK Academy Management System (AAMS). The diagram includes entities, attributes, relationships, primary keys, and Crow's Foot cardinalities.

REFERENCES

1. R. Elmasri and S. Navathe, *Fundamentals of Database Systems*, 7th ed. Pearson, 2016.
2. Lucidchart ER Diagram Tool. Available: <https://www.lucidchart.com>
3. Chatgpt , Google

APPENDIX: AI USAGE AND PROMPTS

Chatgpt , Claude

Prompt	Purpose
Explain supertype and subtype relationships	Understanding inheritance concepts
Generate business rules for academy system	Documentation support
Create data dictionary for entities	Database documentation